MODULE 5- SYSTEM ADMINISTRATION

I- USER ACCOUNT MANAGEMENT

II- ELEVATING USERS ROLES

III- MONITOR USERS ACTIVITY

IV- SYSTEM UTILITIES UNDER ACCESSORIES

V- PROGRAMS AND SERVICES MANAGEMENT

VI- SYSTEM RESSOURCES MONITORING

VII- WINDOWS EVENTS LOGS

VIII- JOBS AND SCHEDULES

IX- SYSTEM MAINTENANCE

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XI- SERVR MANAGER DASHBOARD

XII- INSTALLING AND UNINSTALLING PROGRAMS

XIII- CHECK SYSTEM HARDWARE

XIV- WINDOWS SHORT-CUT KEYS

1- USER ACCOUNT MANAGEMENT

Every machine needs an user account:

- Windows= administrator
- Linux= root

There are 2 types of users account that are created on an operating system

- The local account (local account meaning its just yours and it's not connected to others computers or any other authenticated server)
- Directory Service account which allows users to be created at the server level then the user will be log in through that active directory authentication

User Account Management

- What we will learn in this lesson:
- Add a user account
- Remove a user account
- Change the display name of the user account
- Activate the user account
- Deactivate user account
- Understand user accounts.
- View user accounts

11- Elevating User Roles

- Every user in windows has a group
- By default a user created without a group will have users group.

We will learn

- Give regular users administrative right
- Give users access to filesystem

III- MONITOR USERS

- It is critical for every system administrator to monitor each user activity

We will learn:

- ➢ How to monitor users
- ➢ How to see their activity
- ➢ Manage their running tasks

IV- SYSTEM UTILITY UNDER ACCESSORIES

We will learn…

- Accessing the accessory
- Calculator
- Browser
- Notepad
- Paint
- Remote Desktop Connection
- Spinning tool
- Steps Recorder
- Windows Media Player
- Windows Server Backup
- WordPad

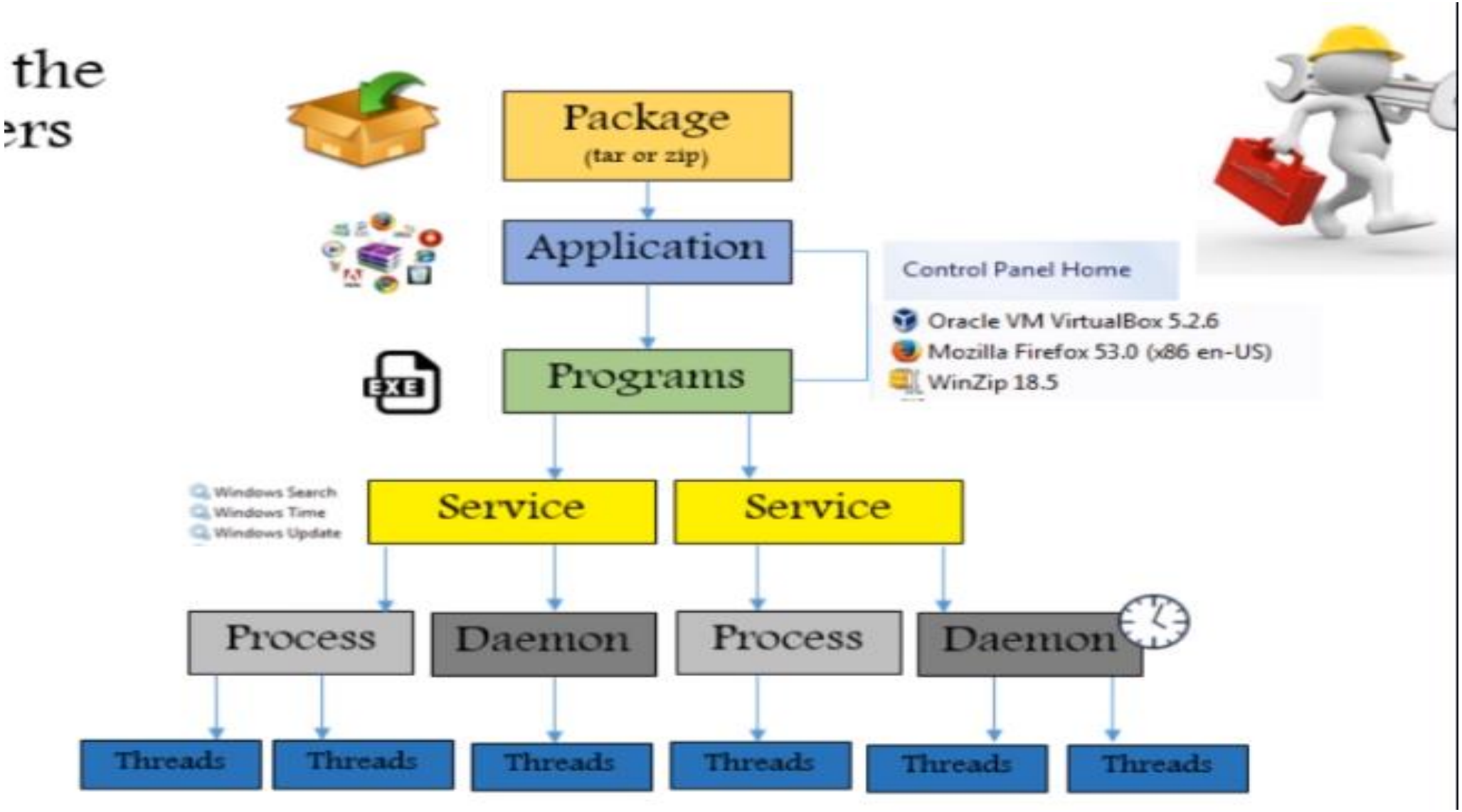
V- PROGRAM AND SERVICE MANAGEMENT

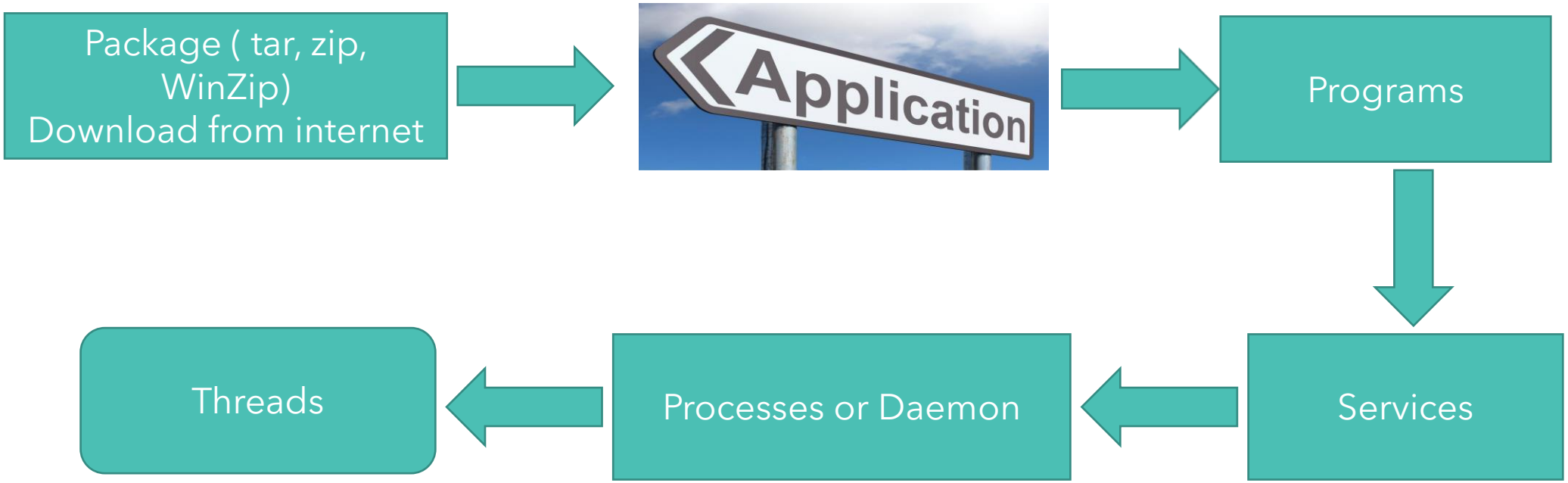
Programs or Applications are the main reasons we use computers.

What is…..

- Package ? Something that is put together into a file and that file is usually like a tar or zip file you can download it from the internet or copy it somewhere
- Application ? Is a computer program designed to carry out a specific task other than one relating to the computer itself, typically to be used by en-users.
- Program ? Is a set of instructions that a computer uses to perform **a specific function.**
- Service ? **Service** is software that performs automated tasks, responds to hardware events, or listens for data requests from other software
- Daemon ? **Daemon** is a process that runs in the background and performs a specific function ... It exists for the purpose of handling periodic service requests that a **computer**.
- Process ? **Process** is simply a program in execution. One program strictly has one process associated with it.
- Threads ? **Threads** are a way for a program to divide (termed "split") itself into two or more simultaneously (or pseudo-simultaneously) running tasks. ... **Threads** are lightweight, in terms of the system resources they consume, as compared with processes.

Program and service management




Package (tar, zip, WinZip)
Download from internet
Application
Programs
Services
Processes or Daemon
Threads

What we will learn

- How to install a program/ Application
- Delete an application
- Package information
- Accessing services
- Managing services (meaning stop, start, and restart)
- Listing processes

VI- SYSTEM RESOURCE MONITORING

- What are system resources? Anything that you need your system to run an application
- We will learn…
- How to access resources monitoring tool.
- Identify intensive processes utilizing high resources.
- Manage resources

VII- WINDOWS EVENT LOGS

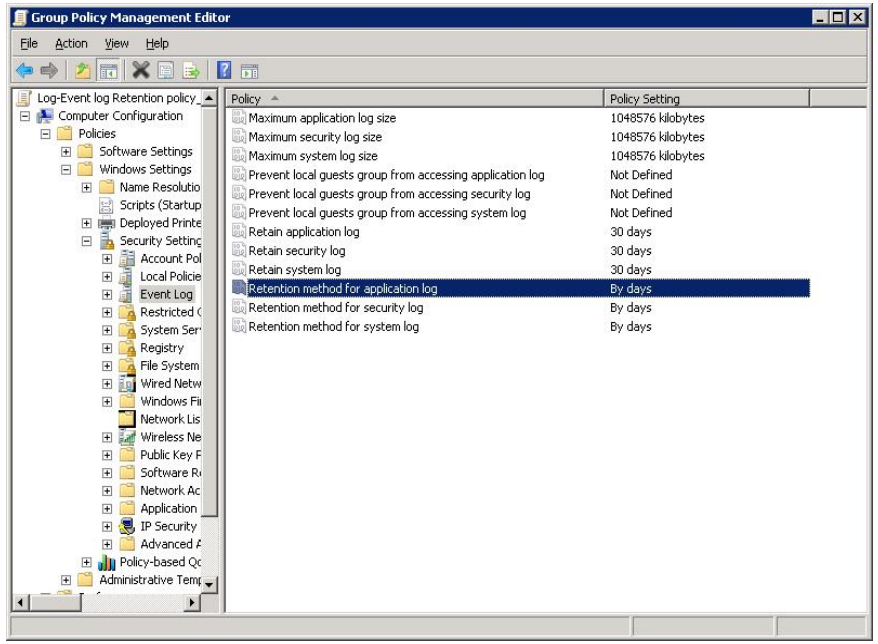
- What are logs?

In computing, a log file is a file that records either events that occur in an operating system or other software runs, or messages between different users of a communication software.

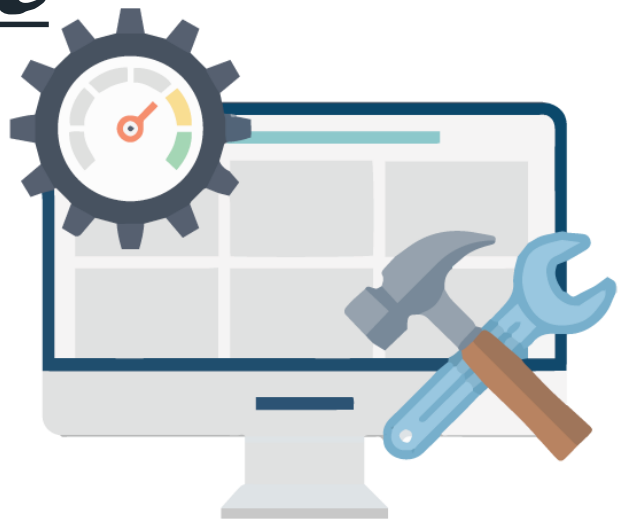
- types of logs:
- Hardware
- Operating system (security, setup, etc.)
- Application

Logs location:

C:\Windows\System32\Winevt\Logs



VII- SYSTEM MAINTENANCE



System maintenance tasks includes:

- Shutdown
- Reboot
- Reboot in single user mode (safe Mode)
- Windows updates
- Patch management (Hotfix)
- Disable remote access

VIII- JOBS AND SCHEDULES

- What's a job?

Job is anything created by a user, administrator or system that perform certain tasks, like create a user, run application, run program, reboot the server.

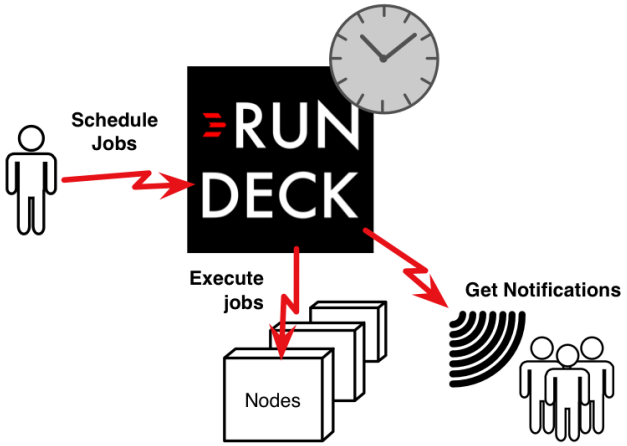
- What is scheduling a job?

We will learn…….

➢ How to create a job (e.g. Reboot the server)

Shutdown /r /t 0

➢ How to schedule that job through task scheduler?



IX- WINDOWS SETTINGS

- Windows Settings is an updated version of control panel.

In windows settings we will learn……

System

Devices

Network & Internet

Personalization

Accounts

Time & Language

Easy of access

Privacy

Update and security

X- SERVER MANAGER

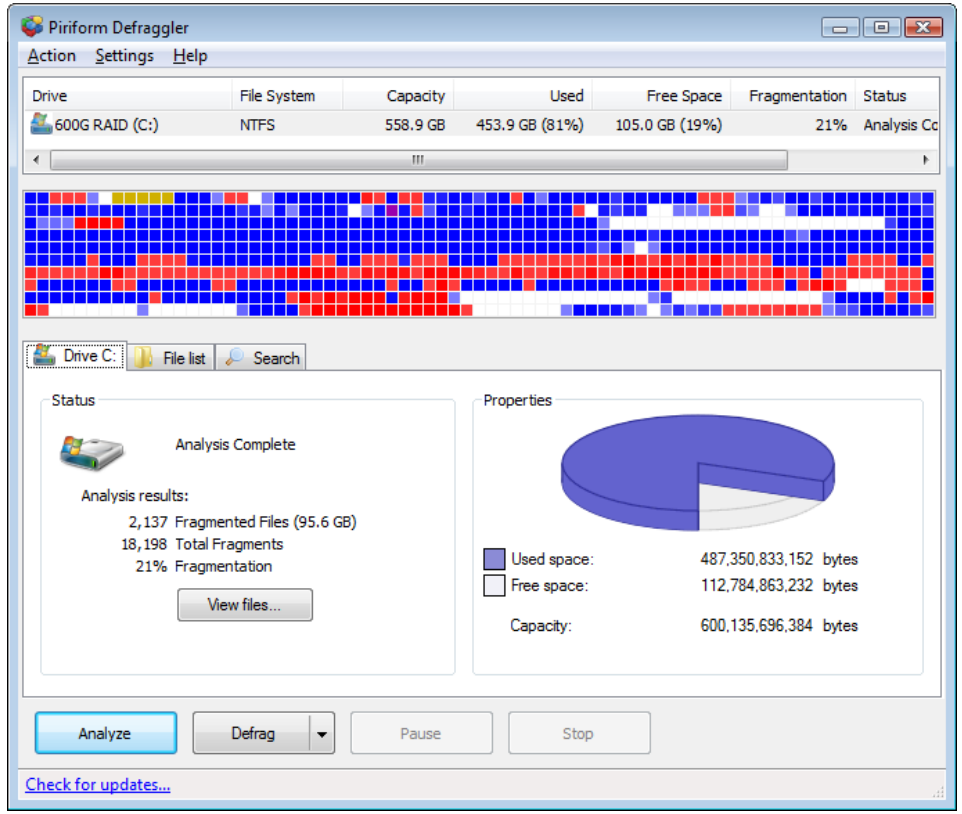
- Server Manager is one of the administrator tool to perform many tasks such as,
- Add roles
- Add and remove features
- Local server properties
- Events
- Services
- Analyzer
- Tools menu

What we will learn…….

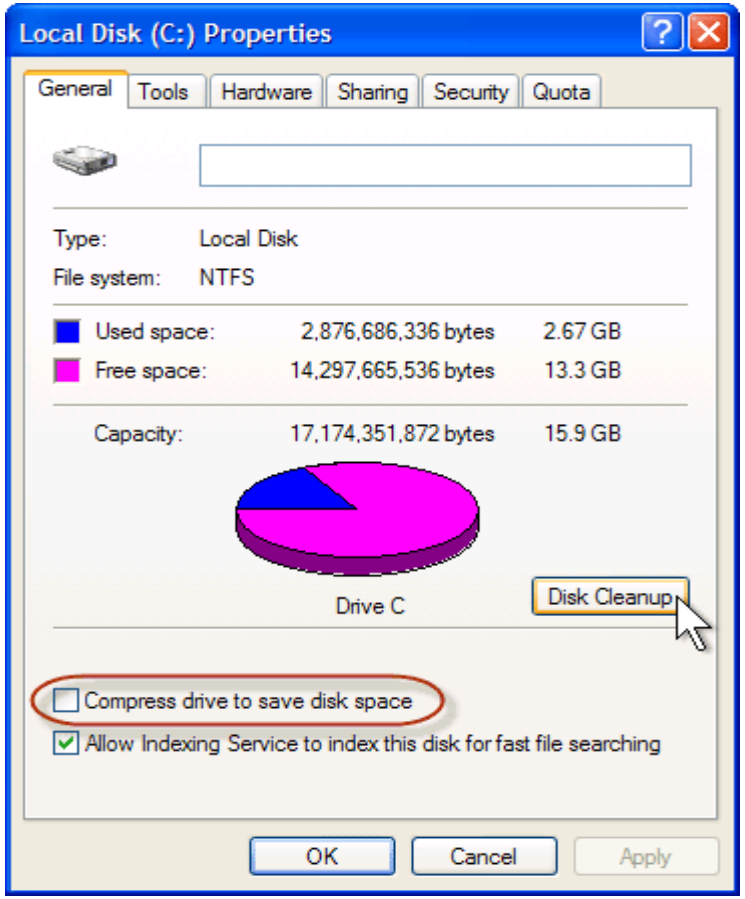
- Everything about server manager and all the features it has to offer.

Server manager

- **Disk defragmentation** takes all the pieces of each file, and stores them in one place. It also makes sure programs are each in one place, and that unused space on the hard **disk** is all together.
- **Disk Clean-up** is a computer maintenance utility included in Microsoft Windows designed to free up disk space on a computer's hard drive.



Defragmentation



Disk cleanup

XI- INSTALLING AND UN-INSTALLING PROGRAMS

- A system administrator has to install, update or re-install programs all the time.
- Reason behind program management

➢ Programs are needed by the users

➢ Updates are installed when newer version is available

➢ Un-install when a program is corrupted or no longer needed.

What we will learn….

❖ Download a package or program

❖ Install

❖ Configure

❖ Manage

XII- WINDOWS APPLICTATIONS

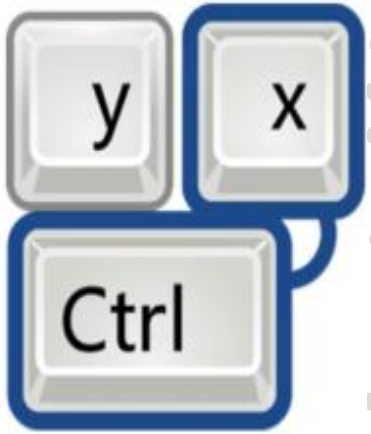
- Some of the important native Windows Applications are:
- Microsoft Domain Controller / Active Directory
- Microsoft DNS or exchange
- All the apps under accessories
- Internet explorer
- Microsoft office
- Skype

Some 3rd party applications are:

- Firefox or Google chrome
- Adobe reader
- Oracle VirtualBox
- VMware vSphere

XII- Windows short-cut key

Ctrl + Alt + arrow keys	Change the entire screen display
F1 key	Get help on a selected command
F5	Refresh
Alt+Ctl+Del	Main system operation
Windows Key + R	Opens the Run menu
Windows Key + E	Opens filesystem explorer
Ctrl + Shift + Esc	Open Task Manager
Windows Key + Break	Opens system properties
Windows Key + F	Opens search for files and folders
Windows Key + D	Hide/display the desktop
Alt + Esc	Switch between programs in order they were opened
Windows + Tab	Open Task view

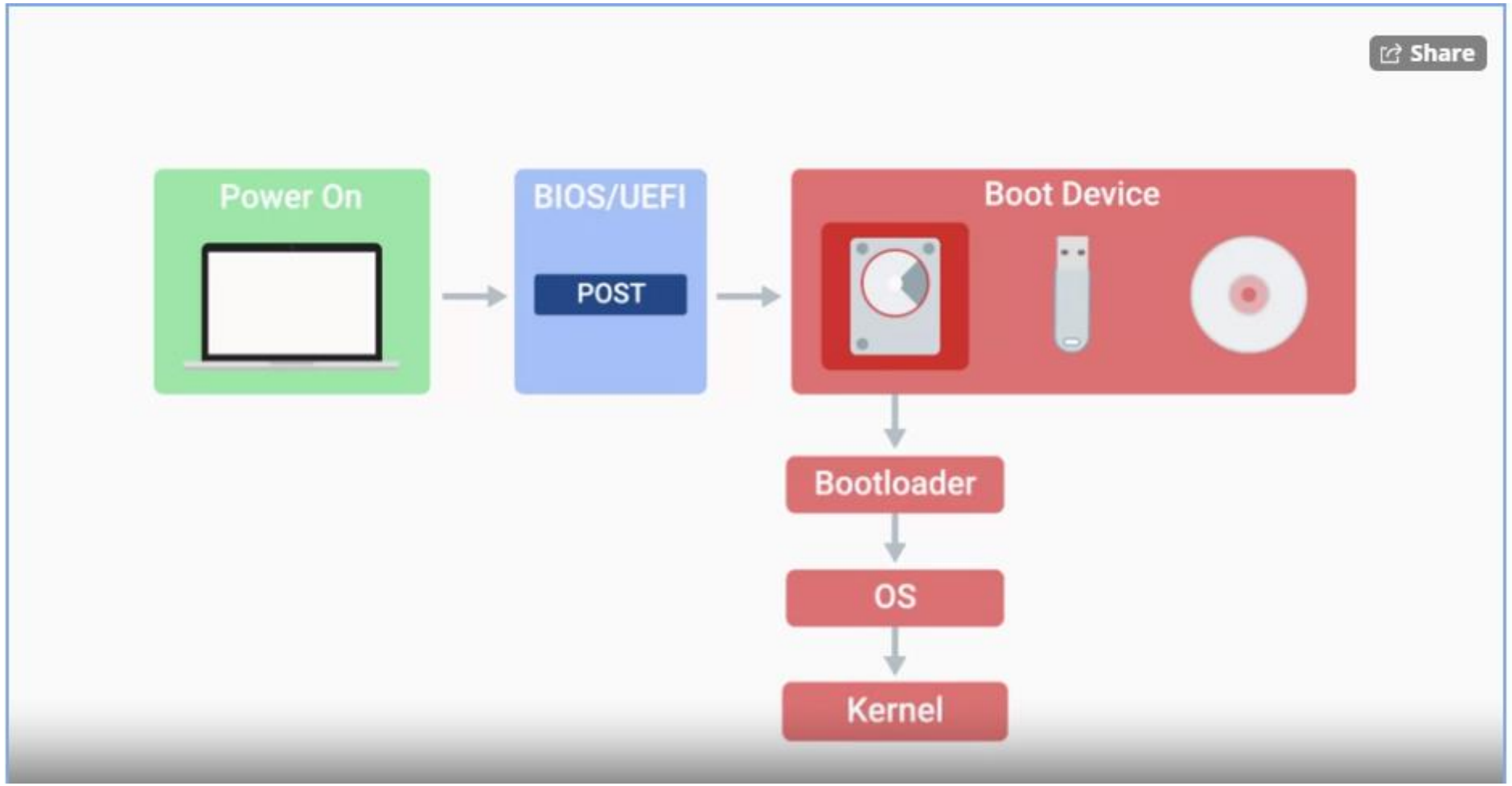


XIII- System Hardware (Device Manager)

- Hardware is anything inside of your computer or attached to your computer.
- A computer can have hardware from different manufacturer.
- Knowing your hardware is important to troubleshoot issues

The Boot Process

- **Phase 1 – Preboot**
- In this phase, the PC's firmware is in charge and initiates a POST and loads the firmware settings. Once all this works (hopefully), the system identifies a valid system disk and reads the MBR. The system then starts the Windows Boot Manager. This is located here: %SystemDrive%\bootmgr
- **Phase 2 – Windows Boot Manager** This step determines if you have multiple OS installed on your computer. If yes, then it offers a menu with the names of the OSs. When you select the OS, it will load the right program, i.e., Winload.exe to boot you into the correct OS
- **Phase 3 – Windows Operating System Loader** Like its name, *WinLoad.exe* loads important drivers to kick start the Windows Kernel. The kernel uses the drivers to talk to the hardware and do rest of the things required for the boot process to continue
- **Phase 4 – Windows NT OS Kernel :** This is the last stage that picks up the Registry settings, additional drivers, etc. Once that has been read, the control is taken by the system manager process. It loads up the UI, the rest of the hardware and software. That's when you finally get to see your Windows 10 Login screen


Share
Power On
BIOS/UEFI
POST
Boot Device
Bootloader
OS
Kernel